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Project lifeline: implementing SBIRT in rural pharmacies to address opioid overdoses and substance use disorder

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ABSTRACT

Background: There is emerging recognition of the unique benefits of implementing screening, brief intervention, and referral to treatment (SBIRT) in pharmacy settings to identify patients who can benefit from services and connecting them to those services.

Objectives: This study describes Project Lifeline – a multipronged public health initiative to provide educational and technical support to rural community pharmacies implementing SBIRT for substance use disorder (SUD) and providing harm reduction support.

Methods: Eight community pharmacies were recruited. Patients receiving a Schedule II prescription were invited to engage in SBIRT and offered naloxone. Patient screening data and key informant interviews with pharmacy staff on implementation strategy were analyzed.

Results: Between 2018–2020, 4,601 adult patients were offered screens and 3,407 screens were completed on 2,881 unique adult patients (51.3% female; <0.01% nonbinary; 95.7% White). Of these unique screens, 107 patients were indicated for brief intervention, 31 accepted the brief intervention; and 12 were given a referral to SUD treatment. Patients who declined SBIRT or who did not want to reduce their use were offered access to naloxone ($n = 372$). Key informant interviews highlighted the importance of person-centered staff education, role-playing, anti-stigma training, and integrating activities into existing patient-care processes.

Conclusion: While ongoing research is needed to characterize the full impact of Project Lifeline on patient outcomes, the reported findings help reinforce the benefits of multipronged public health initiatives that include community pharmacists to address the SUD crisis.

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Introduction

Non-medical prescription opioid use and opioid use disorder (OUD) are substantial public health concerns. In 2017, The Centers for Disease Control and Prevention (CDC) and Substance Abuse and Mental Health Services Administration (SAMHSA) estimated that 1.5 million American adults engaged in non-medical prescription opioid use and more than 650,000 used heroin or illicitly-obtained fentanyl (1,2,). In the same year, 2.1 million Americans met the Diagnostic and Statistical Manual, 5th Edition's criteria for OUD, defined as "a problematic pattern of opioid use leading to clinically significant impairment or distress (3). " Even in the absence of OUD, non-medical prescription opioid use is concerning because of the high risk of overdose death (4). In 2017, opioid overdoses led to 47,600 deaths in the U.S. and 3,705 in Pennsylvania (5,6), which was the third highest opioid overdose death rate in the US overall, and first highest among males ages 25–34 (7–9). As of 2021, these rates have only increased, with 80,816 estimated opioid related deaths in the US (10).

Screening, brief intervention, and referral to treatment (SBIRT) is a comprehensive and integrated public health approach that has been implemented in a variety of healthcare settings including primary care offices and emergency departments (11,12). SBIRT can identify patients along the substance use disorder (SUD) continuum (screen), then provide brief, immediate interventions and appropriate referrals to treatment utilizing warm handoffs (11,13,14). The method allows healthcare providers to assess the severity of substance use, identify appropriate interventions, provide brief advice focusing on awareness regarding substance use and motivation toward behavioral change, and connect patients to specialty care if appropriate. Prior research has highlighted the widespread implementation of SBIRT with multidisciplinary teams as one of the many critical components toward developing a cascade of care to address the opioid epidemic (15,16). Typically, SBIRT has been implemented in primary care (17,18)and emergency department/hospital settings (19) by

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physicians, nurses, and social workers with promising reductions in healthcare costs and utilization (16,18–21).

There is emerging recognition of the unique benefits of implementing SBIRT in diverse settings, including pharmacies (16,22). Pharmacists monitor prescriptions for contraindications, help prevent medical errors, and provide patient and provider education (23). Pharmacists are poised for diminishing opioid-related harms by having the final scheduled face-to-face interactions with patients at their highest risk moments (i.e., at point of prescription refill) and having more patient interactions overall relative to primary care providers (23). Patients positively screened at the point of prescription fills may be directly linked to SUD treatments and provided critical harm reduction resources such as syringe exchange, hepatitis B vaccination, and naloxone distribution (24).

Project Lifeline is a 15-month initiative educating pharmacists and pharmacy staff and supporting community pharmacies in implementing SBIRT services into their patient-care processes. Project Lifeline emerged as an ongoing collaboration between the implementation team and the Blair County Single County Authority (SCA) – a state appointed, nonprofit corporation that provides community assessment, capacity building, planning, implementation and evaluation to support drug and alcohol programming for the county (25). The implementation team then worked with the leadership of two local networks of independent and corporate-affiliated pharmacies to identify and recruit eight sites based on patient volumes. When Project Lifeline began, Blair County had 34 pharmacies serving an estimated population of 124,736 people (94.1% Non-Hispanic White, 2.2% Non-Hispanic Black, 1.4% Hispanic) in an area of 527 square miles (26). In this paper, we describe the Project Lifeline Systems Transformation Framework (STF) implementation strategy, SBIRT adoption, fidelity, as well as perceived impact, barriers, and facilitators.

Methods

Project Lifeline aimed to (1): implement screening with patients receiving opioids for SUD risk (2); reduce patient risk for opioid overdose via naloxone dispensing; and (3) coordinate care with community partners involved in the treatment continuum. The University of Pittsburgh Institutional Review Board deemed Project Lifeline as non-human research. Please see Figure 1 for an overview of Project Lifeline implementation.

Over the 15-month initiative, eight pharmacies in Blair County, Pennsylvania participated in Project Lifeline including two regional grocery chain pharmacies, four regional traditional chain pharmacies, and two independently-owned pharmacies. Each pharmacy implemented SBIRT, and inclusion criteria for patients receiving SBIRT services included being age 18 years or over and picking up or dropping off a Schedule II prescription (opioid or stimulant). Each patient with an opioid prescription was offered naloxone. Exclusion criteria for receiving SBIRT services included being aged less than 18 years, a third-party individual picking up a prescription on behalf of someone else (e.g., friend/family member), and patients receiving their prescription via delivery. Pharmacists asked all eligible patients for permission to provide SBIRT services, and patients were given the right to refuse any or all services.

	Project Lifeline Phase		
	1	2	3
	Months 1-6 (Jan-Jun 2018)	**Months 7-21 (Jul 2018-Sep 2019)**	**Months 22-26 (Oct 2019-Feb 2020)**
Milestone	• Assemble Executive Committee and hold first weekly meeting by April 2018. • Complete pharmacy and Site Champion recruitment • Complete Organizational Health Assessments, synthesize results to develop protocols • Complete training plans and protocol development • Begin initial online SBIRT and Naloxone trainings	• Implement SBIRT protocols at each pharmacy and begin collecting screening data by July 2018. • Conduct first monthly site visits by July 2018. • Continue monitoring for fidelity and providing TA • Complete initial online SBIRT and Naloxone trainings by Sep 2018. • Continue SBIRT, Naloxone and other harm reduction trainings/ providing educational materials	• Complete project sponsored data collection for screenings by Dec 2019 • Complete KIIs, conduct final data analyses, and compile final report by February 2020 • Complete project sponsored trainings and TA

Figure 1. Project lifeline implementation timeline. *Note*. KII = Key Informant Interviews. SBIRT = Screening, Brief Intervention, Referral to Treatment; TA = Technical Assistance

SBIRT implementation and fidelity

All staff working in the pharmacy who had patient contact were trained in SBIRT delivery, including pharmacists, pharmacy technicians, and cashiers. Training all staff with potential patient contact was intended to maximize screening potential, regardless of the staffing and patient care models. Together, the initial and full screening lasted a maximum of 10 minutes, though a screening typically lasted 3–5 minutes. Patients were not charged for SBIRT services.

Screening was initially done using paper forms, but two months into the project, screening transitioned to an electronic tablet. Paper copies of the screen were still made available to drive-through patients and patients unable to complete the survey electronically. The screen consisted of an initial patient demographic section and a three-question substance use triage section (i.e. a single-question screen for alcohol use (27), a single-question screen for illicit and prescription drug use (28), and a single-question on family and friends' substance use history). Patients screening positive (responded with any positive endorsement) on the initial questions completed full screens for SUD risk (described in Measures below). The Project Lifeline web-based application then prompted pharmacists to offer patients one of three interventions corresponding to their risk score (1): positive feedback on their screening results (2); a brief intervention (a 3- to 5-minute conversation with the pharmacist in which motivational interviewing (MI) was used to address substance use and discuss harm reduction measures); or (3) a warm handoff to a local drug and alcohol agency for a level of care assessment and connection to services. In a warm handoff, patients who screened positive on a screening tool were immediately introduced to a specialty care provider for services. This occurred through a 24/7 Project Lifeline hotline staffed by case managers who could meet patients at an agreed-upon location or through in-person connections at the pharmacy as certified recovery specialists (CRS) rotated to different pharmacies weekly.

SBIRT fidelity/performance measures included screening rate (i.e., percent of eligible patients screened for SUD), SBIRT score distribution (i.e., percent of patients scoring in each SBIRT risk category), SBIRT service provision (i.e., percent of eligible patients provided with brief interventions and/or referred to treatment), and naloxone units dispensed. During monthly site visits, implementation staff shadowed the patient-care practice workflow, identified process gaps, and provided feedback to pharmacy staff in real-time.

Naloxone and harm reduction service implementation

In Project Lifeline, pharmacy staff were encouraged to provide all patients filling opioid prescriptions with naloxone medications and Hep B immunizations, to facilitate access to HIV and Hep C testing, as well as to provide corresponding educational resources. These services were not covered by the project but were covered by insurance providers (for insured patients) or the SCA (for uninsured patients) so patients did not incur any out-of-pocket expenses. Out of six payers associated with the Project Lifeline pharmacies, three covered naloxone at 100% cost and another three generated a loss for the pharmacy. Aggregate levels of naloxone distribution were tracked by pharmacy staff in computer reports that were then shared with implementation staff during weekly calls.

The Systems Transformation Framework (STF)

Please see Figure 1 for an overview of Project Lifeline implementation and Figure 2 for an overview of the SBIRT workflow. Project Lifeline implementation was guided by the STF, which conceptualizes organizations as complex adaptive systems (29) consisting of several key domains including vision, structure, culture, behaviors, and performance measurements (30). The STF has been used to implement various healthcare innovations of similar scope within pharmacy (31,32) and other healthcare settings. First, Site Champions were identified and asked to complete organizational health assessments to measure their pharmacy's readiness for implementing Project Lifeline, based on the STF domains of Vision (e.g., "I see my role as a leader to be good at communicating to the staff how we can succeed in achieving our vision"), Culture (e.g., "How important is it to your practice to learn new ways of improving patient care?"), Structure (e.g., "How organized would you say the patient care processes are in your practice?"), Behaviors (e.g., "How common is it for staff/providers in your practice to treat each other with dignity and respect?"), and Performance Measurements (e.g., "How often does your practice use any data to improve the services it provides?"). At the same time, implementation staff conducted current condition analyses by observing the patient-care process at each pharmacy. Results from the organizational health assessments and current condition analyses were used to identify potential barriers and facilitators to SBIRT implementation, which informed protocol development and technical assistance needs.

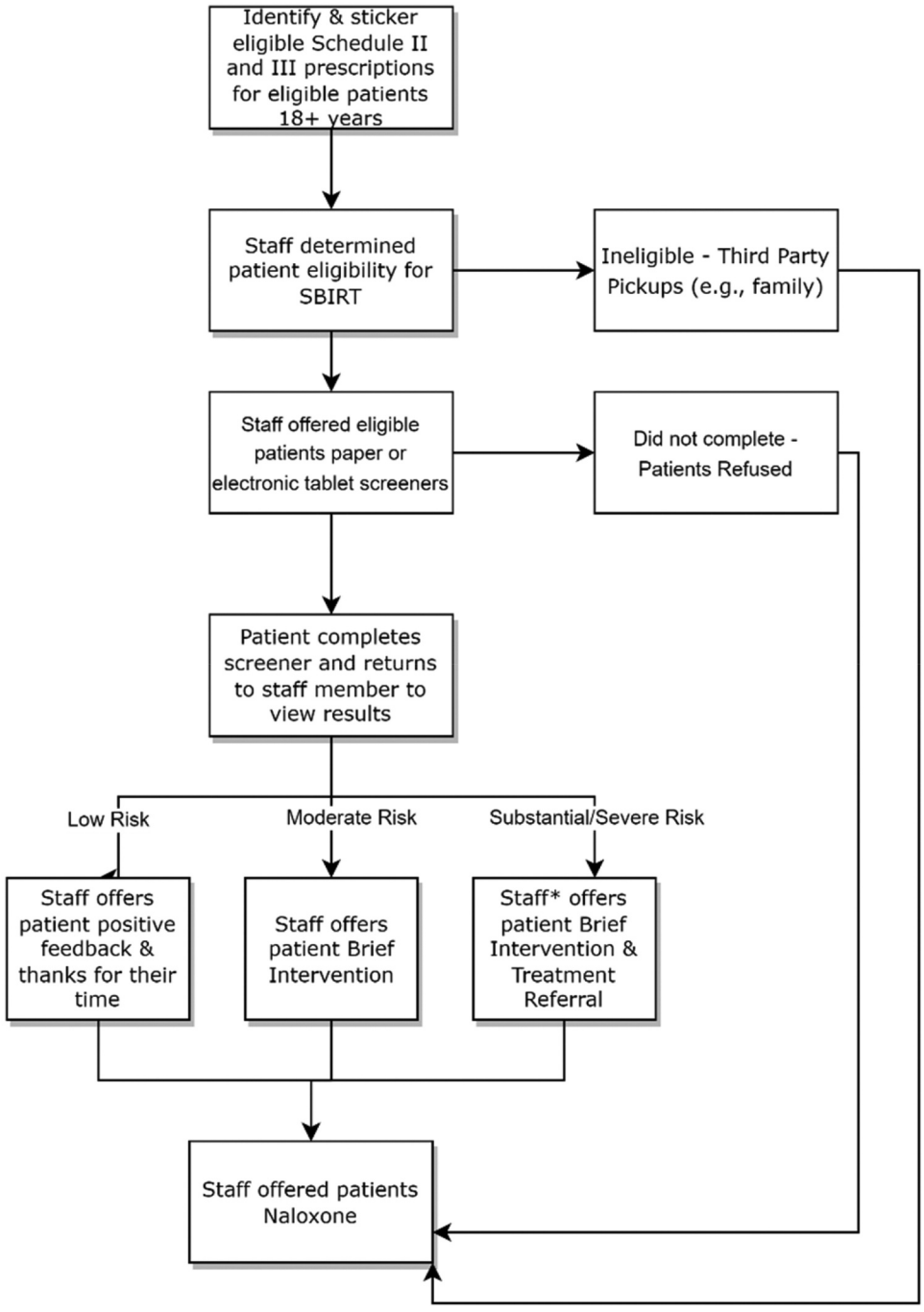


Figure 2. Project lifeline SBIRT workflow. *Note*. Seven months into the project, a Certified Recovery Specialist was embedded into the pharmacies to conduct warm handoffs and increase the rate of referral into treatment.

SBIRT protocols were tailored to the unique needs of each pharmacy, as well as highly specified by timing, sequence, and persons responsible. After protocol development, all Site Champions and other pharmacy staff underwent a series of trainings including an online training course (covering SBIRT, MI principles, pharmacotherapy for OUD, naloxone dispensing, and CDC safe opioid prescribing guidelines) and a 12-hour workshop reinforcing the online training and providing hands-on, interactive practice. Implementation staff would monitor screening and other fidelity rates; if changes didn't occur, they would conduct in person observations and A3s to identify workflow challenges, then provide training until fidelity was reached.

Consistent with prior work, Site Champions were identified staff members who ensured the implementation integrated seamlessly into the clinical workflows at their organization (33). They are distinguished from

other pharmacy staff in their intrinsic interest and commitment to implementing changes as well as their additional effort to lead and oversee implementation activities when implementation staff are not present. Site Champions received additional training and ongoing technical assistance in conducting regular walkthroughs, providing on-site training and related assistance to staff and providers, leading implementation team meetings, and providing ongoing site evaluations. In the current project, Site Champions ensured pharmacy staff were trained in Project Lifeline protocols, answered patient questions, conducted weekly walkthroughs to identify gaps in patient care, reported relevant data in real-time, and participated in weekly technical assistance calls and monthly site visits.

Measures/outcomes

Patients completed two preliminary screener items; if either item was endorsed, they then completed the US Alcohol Use Disorder Identification Test (USAUDIT) and Drug Abuse Screening Test-10 (DAST-10) as full 10-item screens for SUD risk (34,35). Patients scoring 0–7, 8–15, or 16–40 on the USAUDIT or 0, 1–2, or 3–10 on the DAST-10 were categorized as having low, moderate risk, or substantial/severe risk, respectively. Categorical risk scores then indicated whether patients received no action, Brief Intervention (BI), or Referral to Treatment (RT). Though the specific items and numeric scoring systems are unique to each tool, both the USAUDIT and DAST-10 have demonstrated reliability (e.g., α's > .85, test-retest *r*'s >.70) and validated cutoff scores that correctly identify>80% of SUD diagnoses identified via clinical interviews in prior research (36). Both tools are also used by SAMHSA as standard outcomes to guide treatment recommendations (37).

Implementation staff conducted 20–30-minute key informant interviews with seven Champions and three formal organizational leaders at six-month intervals to identify lessons learned and areas for improvement. All participants answered a standard set of 19 questions over the phone. Responses were recorded, transcribed, and compared to identify themes using content analysis by the implementation team. Please see the Appendix for the specific questions asked in the interview.

Results

Patients served and screens completed

From 2018–2020, Project Lifeline served and offered screens to 4,601 patients at eight pharmacies in Blair County, Pennsylvania. Of the patients who received services, 3,407 (74.0%) screens were completed for 2,881 unique patients, representing approximately 2.3% of the total population of Blair County. Reported demographics indicate 48.6% of unique patients were male, 95.7% were White, and 54.8% were aged 55 and older. While patient race was congruent with the demographics of the general population of the County, the study had a larger percentage of males and individuals aged 55+ than expected based on census statistics (38). Table 1 summarizes patient demographics of unique patients screened by the pharmacy.

A total of 3,407 discrete screens were administered; this number is reflective of total SBIRT services provided at each pharmacy and includes patients who completed multiple screens ($n = 526$ screens were repeat screens). Of the total completed screens, 61.1% ($n = 2,082$) were negative, 35.7% ($n = 1,218$) were positive for additional screener questions and negative for a brief intervention, 2.7% ($n = 95$) were positive for a brief intervention, and 0.4% ($n = 12$) were positive for a referral to SUD treatment. Table 2 provides an overview of progress toward project indicators as reported in the Project Lifeline web-based application and Figure 3 presents the screening rates across the project over time.

Patients who did not complete the screen were offered naloxone services and engaged in conversations about opioid medication safety. Of the 1,556

Table 1. Patient demographics by pharmacy.

	Gender			Race		
Site	Male n (%)	Female n (%)	Non-binary n (%)	White n (%)	Non-White n (%)	Mean Age (SD)
Pharmacy A	86 (39.3%)	133 (60.7%)	0 (0.0%)	217 (99.1%)	2 (0.9%)	52.9 (14.91)
Pharmacy B	50 (51.5%)	47 (48.5%)	0 (0.0%)	96 (99.0%)	1 (1.0%)	54.7 (15.16)
Pharmacy C	74 (43.8%)	95 (56.2%)	0 (0.0%)	164 (97.6%)	4 (2.4%)	51.1 (15.0)
Pharmacy D	688 (50.8%)	665 (49.2%)	0 (0.0%)	1256 (95.4%)	61 (4.6%)	55.3 (16.0)
Pharmacy E	90 (38.3%)	145 (61.7%)	0 (0.0%)	223 (96.5%)	8 (3.5%)	54.11 (16.0)
Pharmacy F	130 (46.1%)	151 (53.5%)	0 (0.0%)	269 (95.7%)	12 (4.3%)	52.0 (13.8)
Pharmacy G	82 (53.6%)	70 (45.8%)	1 (0.7%)	141 (92.2%)	12 (7.8%)	58.7 (12.6)
Pharmacy H	175 (54.9%)	144 (45.1%)	0 (0.0%)	298 (94.0%)	19 (6.0%)	57.7 (12.6)
Total	**1375 (48.7%)**	**1450 (51.3%)**	**1 (0.04%)**	**2664 (95.7%)**	**119 (4.3%)**	**54.56 (14.59)**

N = 3,407 patients screened across eight unique pharmacies. Percentages are reported accounting for missing data.

Table 2. SBIRT distribution by pharmacy.

		Screening Rate (%)	Screening and Feedback (%)	BI (%)	RT (%)
Benchmark	Total Number of Screens	100%	(<85%)	(10%)	(5%)
Pharmacy A	251	74%	176 (95.1%)	6 (3.2%)	3 (1.6%)
Pharmacy B	113	100%	111 (98.2%)	2 (1.8%)	0 (0.0%)
Pharmacy C	250	46%	103 (88.8%)	13 (11.2%)	0 (0.0%)
Pharmacy D	1560	17%	236 (89.1%)	28 (10.6%)	1 (0.4%)
Pharmacy E	252	19%	38 (80.9%)	9 (19.1%)	0 (0.0%)
Pharmacy F	369	80%	268 (90.8%)	20 (6.8%)	7 (2.4%)
Pharmacy G	210	61%	126 (98.4%)	2 (1.6%)	0 (0.0%)
Pharmacy H	402	44%	160 (90.9%)	15 (8.5%)	1 (0.6%)

BI = Brief Intervention; RT = Referral to Treatment.

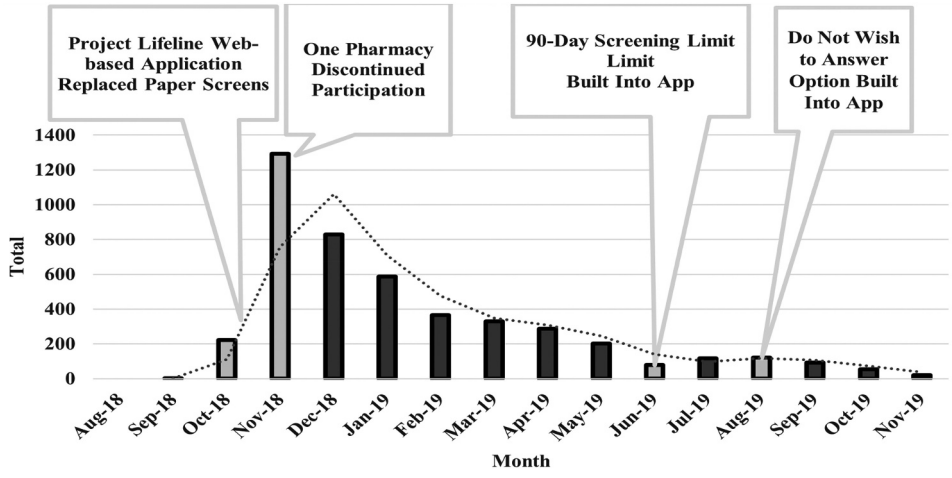


Figure 3. Total patient screens by month over the course of *Project Lifeline*. *Note*. Callouts point to major project changes that occurred. The dotted black line depicts the moving average which is calculated using the subset of data collected up until that timepoint. For example, the moving average of Sep-18 = (Aug-18 + Sep-18)/2; the moving average of Oct-18 = (Aug-18 + Sep-18 + Oct-18)/3, etc.ln

patients who did not complete the screen, 533 provided a reason for refusing SBIRT services. The most common reasons for refusing SBIRT services were previous participation $n = 315$ (59.1%), $n = 99$ lack of time (18.6%), $n = 41$ discomfort with the electronic tablet (7.7%), and $n = 6$ previous enrollments in SUD treatment (1.1%).

SBIRT services provided

From the full screening, 107 patients were indicated for a brief intervention; of which, 12 were indicated for both a brief intervention and referral to treatment. Of the 107 patients indicated, 31 (28.9%) brief interventions were recorded as being completed. Among patients who received a brief intervention, 100% were advised of low-risk guidelines, 74.1% agreed to reduce their use, 6.5% agreed to quit their use, and 90% were already in SUD treatment. Outcome data were missing from one brief intervention. Pharmacists could enter multiple brief intervention outcomes per patient entry.

A total of 9 patients (out of 12) were referred to recovery support services in the community. All 9 (100%) referrals to community services were made via the CRS using the Project Lifeline hotline. Patients were referred to peer support, case management services, support groups, and career services.

The remaining 76 patients declined the brief intervention and 26 provided reasons. The reasons reported for not providing a brief intervention included that the patient refused to discuss (42.3%), the patient left before the pharmacist was notified by a technician/cashier (11.5%), the patient was too busy (3.8%), there was an urgent medical concern (3.8%), and other/non-specified (34.6%).

Project lifeline protocol changes and screening rates over time

There were four significant changes to the Project Lifeline implementation model. In October 2018, the pharmacies switched from using paper screening

forms to electronic tablets. This change was made to improve data collection, storage, and analysis while increasing patient privacy through data encryption. In November 2018, one pharmacy withdrew from the project due to concerns about the time demands for pharmacy staff. In June 2019, about halfway through the project, pharmacies began screening patients every 90 days rather than every time they filled a Schedule II prescription to address staff and patient engagement concerns. Finally, in August 2019 a "Do Not Wish to Answer" option was added to the pre-screen questions. The effects of these changes on the number of screenings completed over time are shown in Figure 3.

Key informant interviews

Key informant interviews were conducted with a Champion at seven of the eight participating pharmacies and with three leadership representatives from the organizations participating in Project Lifeline, for a total of ten interviews. All interviewees were licensed pharmacists in the state of Pennsylvania. Overall, responses revealed numerous common themes. Each theme was expressed by at least two key informants.

Educational opportunity

The first theme is that Project Lifeline *supported patient-focused continuing education* for the pharmacists. One Champion commented, " . . . the whole project has been useful, in a way, because it really helped me open that door. [Substance use] wasn't something that was ever focused on . . . you got me into conversations that I never expected myself to get into – that's very beneficial [for continuing education and patient care]." Another Champion stated, "I think it's been a good change for the positive. I understand the risks associated with alcohol and drugs, but what it's done for me is show me the amazing amount of services that are available to these patients. I think if there's anything SBIRT has done, it's opened my eyes to the tremendous amount of help that's out there for patients." These comments reveal that Project Lifeline changed pharmacists' attitudes and understanding toward working with patients who use substances.

Skill building opportunity

Nearly every interview highlighted *the importance of role-playing* brief interventions for skill development. As one Champion remarked, "I think doing an in-person, face-to-face fake encounter would help everyone, 'cause we did that for me, and it really did help." Further, another Champion commented, "That's how I did most of my training throughout [pharmacy] school, using standardized patients. That's how I got comfortable interviewing and communicating with patients. I think that's a great training tool. I think it's much easier. I think the best experience you get is hands-on experience. You can tell me to do something over and over, but until I'm in there doing it, that's when I succeed." Finally, one member of leadership commented, "There's no substitute for hands-on practice. Lectures are great, but they aren't going to help you talk to the patient in front of you."

Stigma reduction

Responses indicated that Project Lifeline *increased pharmacists' comfort and openness to working with patients who use substances*. One Champion indicated " . . . that the project opens up the conversation about potential drug use between [the] pharmacy and [the] patient . . . I liked that." Another Champion said, "I think it made me a little more comfortable to talk about those things. And that there are those people who realize they have a problem and they do want help. People are willing to talk about it openly . . . and I was surprised by it . . . you know it changed my opinion. It did change how I viewed people. Everyone is a person, and everyone deserves respect."

Implementation facilitators

Further responses revealed the elements that facilitated SBIRT implementation at each site, including tailoring technical assistance and collaborating with pharmacy staff to develop a flexible, highly specified patient care process for each pharmacy. For example, several key informants expressed that the implementation staff were *responsive* to their needs throughout the project. A Champion stated, "you guys are always there to answer any questions we had. There was never a moment where we were left stranded." Most Champions also felt that seamlessly integrating programs into their pharmacy's unique patient-care workflow, including the development of a web-based application, made implementing SBIRT services *more efficient* and *user-friendly* for pharmacy staff and patients. One member of leadership stated, "having the tablets available and using technology. Not just pencil and paper. I think patients are more comfortable with the confidentiality of the tablet because they're not filling out a paper that gets thrown back in the pharmacy somewhere." As one Champion stated, "You [implementation staff] implemented technology into the workplace, and I think that made the process go much more smoothly than worrying about pieces of paper. I think the technology atmosphere was a huge step in that process."

Implementation barriers

In addition to factors that facilitated implementation, interviews revealed challenges that may affect SBIRT implementation, such as long-term financial sustainability [with one response noting *lack of insurance coverage for patients receiving SBIRT services*]; "we need to ensure that . . . the time we put in" and the care delivery model are sustainable. Additionally, *stigma* was reported – including both pharmacy staff assumptions about those who use substances and patient assumptions about pharmacy staff's willingness to support them. "Stigma . . . keeps people from answering the questions real well . . . patients think we're only concerned with the business transaction."

Sustainability

When asked about what aspects of the Project Lifeline and SBIRT protocols the pharmacies intended to continue and end, the majority of informants indicated that they would continue holding the conversations, offering brief interventions, referring patients to treatment services/serving as a connector. As one Champion noted "Having these discussions and knowing when a brief intervention is necessary. . . . And we're dispensing a ton more Narcan, and that's a great touch point for counseling patients on an overdose." Despite placing a high value on the knowledge gained from the conversations yielded by the screenings, eight out of ten informants indicated that they would not continue screening every patient after the project ended because of the burden it placed on staffing/workflow, lack of technical infrastructure, and lack of financial sustainability. For example, one pharmacy staff member noted "We won't be doing SBIRT. We won't be screening because we won't have the iPad. I mean, if we notice something or if someone comes up and reveals that, now we know how to do a brief intervention and refer them to help." In response to the question about what would make it easier for continuing to implement SBIRT, a leader noted "Getting reimbursement. As pharmacy gets tighter and tighter with reimbursement, we need to ensure that there's a return on the investment of the time we put in."

Discussion

Project Lifeline successfully integrated SBIRT into community pharmacies. While there were fewer positive screens and more major barriers to longer term sustainability to continued screenings than expected, Project Lifeline enhanced awareness, knowledge, relationships, and access to critical services (e.g., naloxone, treatment referrals) that can improve the continuity of care for community adults with substance use disorder. Key informant interview findings are consistent with previous research suggesting that while pharmacists and other healthcare providers are open to discussing substance use with patients, they often lack the confidence to do so and are afraid of potentially singling out or offending their patients (39). Providing continuing education and ongoing training to pharmacy staff contributed to increasing staff confidence and changing attitudes, supporting previous findings that education and training increases the likelihood pharmacists will engage in OUD prevention and treatment activities (40). Moreover, implementing a systematic, public health approach to OUD prevention and treatment involving universal screening with validated instruments helped to normalize the conversation, prevented pharmacists from relying on inherent biases, and helped ensure all patients received an appropriate intervention based upon their level of risk.

Project Lifeline was unique from other studies of similar scope in several ways. Previous studies in community pharmacies focused on screening and brief intervention (SBI) for SUD without the referral to treatment component (41,42). A 2019 study from Shonesy et al. implemented a full SBIRT model within a community pharmacy; however, a research assistant, rather than a pharmacist, carried out all of the SBI activities and no warm handoffs were used (43). In contrast, Project Lifeline implemented SBIRT involving the entire pharmacy team (i.e. pharmacists, technicians, cashiers, etc.) and provided active linkage to SUD treatment via warm handoffs using a CRS embedded into the pharmacies. As a result, 75% or $n = 9$ of patients who were not already engaged in SUD treatment and administered a brief intervention agreed to reduce their use or quit altogether. 100% or $n = 9$ of patients referred to recovery support services were referred by CRS using the Project Lifeline hotline. These findings are consistent with literature suggesting facilitated referrals increase access to services when compared to passive referrals (14,44). Thus, Project Lifeline has demonstrated the feasibility of warm handoffs to SUD treatment within community pharmacy settings.

Project Lifeline is also unique in that pharmacists received training in naloxone dispensing and facilitated naloxone access to eligible patients. As such, patients who declined SBIRT or were not ready to reduce their use were given access to life-saving harm-reduction measures known to reduce overdose deaths (45). In the present study, nearly 12 times as many patients were given naloxone ($n = 372$) than were given a brief intervention ($n = 31$). Findings highlight the need for a multi-pronged public health approach to overdose

death prevention. Where SBIRT can effectively identify patients at risk, encourage use reductions, and facilitate treatment referrals, harm reduction medications such as naloxone distributions can help reverse overdoses among those who do not have access to the healthcare system, experience recurrent use, or are not ready to receive SBIRT services. Future projects focused on training pharmacists and facilitating patient access to a range of harm reduction measures (e.g., needle exchange, fentanyl testing strips) are warranted (46–48). Researchers, practitioners, and educators in this area should note that trainings that incorporate interactive face-to-face role-plays, anti-stigma trainings, and flexibly tailored technical assistance are both well received and effective (49).

Throughout Project Lifeline, there were four significant changes to the implementation model that impacted screening rates. While some such as switching from paper screening forms to electronic tablets improved data collection, storage, analysis, and patient privacy, others such as allowing pharmacies to reduce screening to every 90 days rather than every visit, reduced screening rates. Ultimately, there were noted capacity challenges with sustaining consistent screening and referrals to treatment by pharmacy staff, demonstrated by declining screening rates toward the end of the project and provider feedback; future projects will need to consider these issues from the outset. A number of factors affected the uptake of services at each pharmacy and the associated performance measures (such as screening rate). Uptake across pharmacies varied. Though not statistically tested, based on staff observation and staff feedback variability in logistical setups, staffing models, staff capacity, and leadership support appeared to contribute to substantial variability in patient screenings. Further, different patient populations visited each pharmacy resulting in varied consistency of patients completing the screen. While the Implementation Team sought to support all pharmacies with tailored protocols, providing booster trainings, and embedding CRS support, leaders noted that the limited funding/revenue to support SBIRT implementation was a barrier to long-term sustainability. Future SBIRT implementation projects will need to consider the development and execution of a comprehensive business plan that will foster its sustainability.

There were also a few identified barriers as Project Lifeline progressed. These challenges provide valuable lessons for future implementation in public health pharmacy settings. Initially, all patients filling Schedule II prescriptions were screened. However, patient encounters during Project Lifeline suggest that patients filling certain Schedule III medications may also be at risk for substance use. In the future, other programs should consider expanding screening to all opioid patients, including Schedule II and Schedule III medications. Initially, pharmacy staff were intended to conduct all of the treatment referrals and warm handoffs. However, seven months into the project after no treatment referrals had been completed, the implementation team hired and embedded a CRS to facilitate warm handoffs. Thereafter, treatment referrals increased substantially. Future projects will want to consider incorporating and integrating additional staff support from the outset. Because patients may not have access to health insurance coverage and may not always be able to pay out-of-pocket for SBIRT services, continuing delivery of care services after Project Lifeline could be affected without a sustainable SBIRT care delivery financial model. If SBIRT and programs like Project Lifeline are to be made available to more patients, sustainable care delivery models are needed.

Limitations to Project Lifeline include all that normally are inherent in observational comparison studies without a control group (50,51). Limitations to external validity include lack of generalizability due to restriction to eight pharmacies in one rural county in Pennsylvania and a large prevalence of regional chain and independently-owned pharmacies. Lack of longitudinal patient follow-up to determine long-term impact is also a limitation. Limitations to internal validity include the lack of a control group and inconsistent data collection (e.g., changing from paper to app screening formats). Hepatitis C/HIV screening was referred from the pharmacy to another care site due to Pennsylvania laws at that time; however, in other states, these care services could be handled within the pharmacy.

Conclusion

Leveraging community pharmacies for implementing SBIRT and the provision of harm reduction services such as naloxone represents a unique opportunity for building a continuum and cascade of care that reaches more patients in their moments of highest need (15). As community pharmacists increasingly provide more direct patient care services (e.g., immunizations, hepatitis and HIV screening), incorporating SBIRT may improve the quality of clinical care delivered overall through more comprehensive screenings that help identify a greater range of risks for co-occurring needs as well as the associated counseling, community referrals, and service provision. This expansion is particularly valuable for rural communities with fewer healthcare providers and facilities that may rely on pharmacies for a wider range of

services and monitoring (24). Project Lifeline data supports the conclusion that comprehensive SBIRT and naloxone distribution services can be implemented in community pharmacies, as well as the conclusion that targeted trainings can build pharmacists confidence in holding challenging conversations and serving as a community connector to key services/resources.

Together, Project Lifeline activities show promise as a multipronged approach to reducing opioid overdose deaths and building a stronger pipeline for a more continuous cascade of care for SUD (15). Project Lifeline also provided insights to facilitators and barriers of SBIRT implementation and longer-term sustainability in community pharmacies that will need to be addressed in future projects, including stigma reduction, financial sustainability for screening activities, and integrating sustainable clinical workflows. While these factors are not unique to community pharmacies, community pharmacies remain an underutilized and under-researched setting for patient outreach; future research focused on the unique strengths and challenges is needed.

Disclosure statement

Joseph Weidman is currently employed by Johnson & Johnson, the majority of his contributions occurred while a student at The University of Pittsburgh and Johnson & Johnson had no role in the study design, collection, analysis or interpretation of the data, writing the manuscript, or the decision to submit the paper for publication.

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